

Simeon Tran

🏠 simeonat.github.io | 📧 [SimeonAT](#) | 🌐 [simeon-tran](#)

Education

University of California, Irvine

PH.D. IN NETWORKED SYSTEMS

Irvine, California

Jun. 2026 - PRESENT

University of California, Santa Cruz

M.S. IN COMPUTER SCIENCE AND ENGINEERING

Santa Cruz, California

Sept. 2023 - Dec. 2025

B.S. IN COMPUTER SCIENCE

Jul. 2019 - Jun. 2023

Master's Thesis

Comparative Study of Encryption Algorithms in Battery Powered Thread® Networks for Smart Homes

Jan. 2024 - Dec. 2025

[HTTPS://SIMEONAT.GITHUB.IO/RESEARCH/THESIS](https://simeonat.github.io/research/thesis)

Poster Abstract Published in SenSys 2026

- Forked and modified OpenThread and ESP-IDF to enable ESP32 802.15.4 compatible hardware to support ASCON lightweight encryption scheme when running OpenThread
- Designed and conducted comparative study to analyze impacts of energy consumption and network performance of authenticated encryption on resource-constrained smart home devices
- Built automated testbed to evaluate network performance and conducted manual energy consumption experiments
- Results show no significant impact on battery lifetime or network performance when ASCON is used instead of AES-CCM

Experience

Sphinox Inc.

Jun. 2022 - PRESENT

FULL STACK WEB DEVELOPER & NETWORK ADMINISTRATOR

- Owned development and long-term operation of Next.js CRM for barber shop customer rewards program containing more than 6000 user records
- Maintained company-wide networking infrastructure containing 5 servers and 20+ devices using VLANs, Tailscale ACLs, and administering UniFi networking equipment, securing internal services and production systems
- Built observability into server infrastructure using Prometheus and Grafana for 24/7 monitoring of uptime, performance, and failures

Tech4Good Lab

Jun. 2023 - Dec. 2023

LEAD RESEARCH ASSISTANT

Santa Cruz, California

- Led team of 4 other students prototype to develop Express.js and OpenAI API backend service integrating ChatGPT to generate personalized learning curricula
- Supported team members new to backend development by assisting in defining backend interface using OpenAPI specification and taught them how to write unit tests using Jasmine testing framework
- Designed and conducted 3 user interviews to evaluate system effectiveness and inform iteration on backend and UX design

WEB DEVELOPER TEAM LEAD

Apr. 2023 - Jun. 2023

- Built cross-platform mobile deployment infrastructure using Ionic to enable preexisting Angular web applications to run on iOS and Android
- Integrated Google OAuth authentication in mobile applications using Capacitor Ionic plugin
- Wrote onboarding instructions and 44-page documentation to help new members to get acquainted with Ionic

Skills

Embedded Systems ESP-IDF, OpenThread, nRF Connect, Analog Discovery 3

Networks TCP/IP, Wireshark, Prometheus, Grafana, UniFi

Web Frameworks Node.js, Deno, Next.js, Astro

Front-end React, Angular, Vue.js

Databases PostgreSQL, Firebase

Back-end OpenAPI, GraphQL, Express.js

CI/CD Docker, GitHub Actions

Programming C/C++, Python, Typescript/Javascript, Haskell

Soft Skills Teamwork, Leadership, Tutoring, Mentoring

Languages English, Vietnamese

Projects

Daily Time Manager Web Application

Sept. 2022 - Dec. 2022

[HTTPS://GITHUB.COM/SIMEONAT/BAGEL.IO/TREE/MAIN](https://github.com/SIMEONAT/BAGEL.IO/TREE/MAIN)

- Worked with four fellow software engineering classmates to create a web app that generates visualizations based on completed tasks on a user's to-do list
- Designed the UI for the task cards and to-do list
- Developed both the React front-end and Node.js back-end implementations for the login and register pages

Fractal Terrain Generation Program for Python Course

Apr. 2020

[HTTPS://USERS.SOE.UCSC.EDU/~PANG/30/S20/PROG1/TERRAIN_SIMEON.TXT](https://users.soe.ucsc.edu/~pang/30/s20/PROG1/TERRAIN_SIMEON.TXT)

- Used Python Turtle Module and starter code provided by the class TA to create program that recursively generates random terrain
- Developed extra features beyond what was asked, which encouraged graders and course faculty to label this program as a “star solution” on course website (<https://users.soe.ucsc.edu/~pang/30/s20/>)

Video Game Demo using Python and PyGame

Dec. 2020 - Dec. 2021

[HTTPS://GITHUB.COM/SIMEONAT/SPACEVIEWERPYGAME](https://github.com/SIMEONAT/SPACEVIEWERPYGAME)

- Created single-player game demo where players interact with objects in outer space
- Implemented RPG-style text dialogue using object-oriented programming
- Applied knowledge of data structures by creating a binary tree to store RPG text-box objects
- Integrated pixel art sprites with Python code to create lively animations

Teaching

Baskin Engineering at UCSC

Jun. 2024 - Jul. 2024

FULL STACK WEB DEVELOPMENT TEACHING ASSISTANT

Remote

- Held daily office hours where I helped students with debugging
- Taught students web development using React, Express, and PostgreSQL
- Helped students develop the habit of Test Driven Development (TDD)

UC Santa Cruz Learning Support Services

Sep. 2020 - Jun. 2021

PYTHON COURSE TUTOR

Remote

- Held weekly remote tutoring sessions with diverse students for Introduction to Python course
- Assisted students on Python concepts and basic object-oriented programming to help them perform well on assignments and exams
- Demonstrated knowledge of Python by creating programming examples and activities

UC Santa Cruz Academic Excellence Program

Oct. 2021 - Dec. 2021

PRECALCULUS TUTOR

Remote

- Was a tutor/mentor for Professor Morales-Almazan's Precalculus class, under the recommendation of the professor himself
- Helped students with Precalculus topics that they struggled with, allowing them to be better prepared for the homework and exams
- Gave freshman students from different backgrounds advice on how to navigate the university, in addition to assisting them with math